12 Friction as a Force

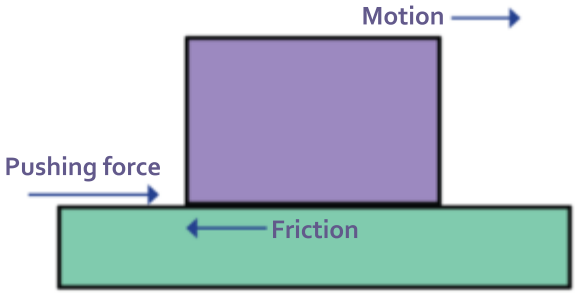
Why does a ball rolling on a floor stop after some time? This happens because of a force called friction. Here, the ball and the floor are in contact. Any force that acts at the point of contact between the two objects is a contact force. Friction is an example of a contact force. Let us learn more about friction.

Learn about

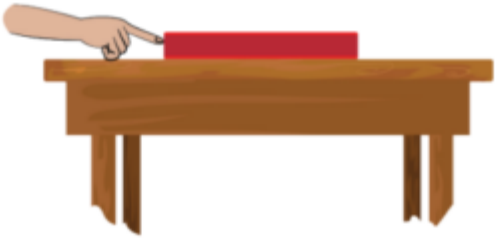
- Friction
- How to reduce friction?
- Uses of friction
- Negative effects of friction

▶ Friction

The force that acts between a moving object and a surface, which either slows down the object or stops it completely, is called the ***force of friction****.*



Let us observe the force due to friction by sliding a book on the table. When we push the book, it moves a certain distance and then stops. This happens because **frictional force** acts between the book and the table. This force always opposes the force that causes the movement. It also acts in the direction opposite to the motion of the book. This force stops the book from sliding further.



Frictional force between the book and the table

Friction is very useful for us. Without friction, an object in motion would never stop. For example, a moving car is able to stop only due to friction. However, if there is too much friction, nothing will move. Materials with a smooth surface, such as ice and glass, have less friction than objects with a rough surface, such as a carpet.

Think and Discuss

Imagine yourself walking on a polished floor and a concrete floor. On which floor will you be able to walk properly if you wear the same type of shoes in both cases?

Activity

Aim: To understand force due to friction

Materials required: A toy car or train, a slope followed by a plain (floor), and a long carpet/mat

Procedure:

1. Place your toy car/train on the slope and let it roll off onto the plain floor (Fig. A).
2. Note the distance the car travels on the floor.
3. Now, cover the slope and the plain floor with a thick carpet.
4. Place the car on the slope again and allow it to roll off (Fig. B). Mark the distance it travels on the floor.

Fig. A

Fig. B

Observation: The toy car will travel a greater distance when the floor is not covered with the carpet.

Conclusion: The floor without the carpet is smooth and, therefore, has less friction.

How to Reduce Friction

Friction comes into effect when two objects are in contact with each other. The smoother their surfaces, the lesser is the force due to friction.

Friction can be reduced by many ways. Some of the ways are as follows:

- **Lubricants** are substances that can smoothen the two surfaces that are in contact with each other, thereby reducing friction. For example, **oil**. We apply oil on the door hinges to reduce the creaking noise while opening or closing a door.
- **Powder** is also used to reduce friction. For example, we apply powder on the carrom board before playing. This makes the movement of the striker and the coins very smooth.

Skating is easier on ice than on grass as ice has less friction.

- **Wax and grease** are also used to reduce friction. Grease, when applied on the moving parts of engines (like that of cars and electric fans), reduces friction.

Activity

Aim: To observe the effect of greasing on a surface

Materials required: Two steel glasses, water, face cream or petroleum jelly

Procedure:

1. Take a steel glass and grease the outer surface by applying cream or petroleum jelly.
2. Wash your hands and dry them.
3. Lift up the first glass that does not have any cream on its surface.
4. After this, try lifting up the glass that has been greased.

Observation: You will observe that the glass without greasing is easier to hold, whereas, the greased glass is slippery and difficult to hold.

Conclusion: The cream reduces the friction between our fingers and the surface of the glass by making it slippery. This is why the greased glass is more difficult to hold.

Uses of Friction

Friction is involved in our day-to-day life. It is useful for us in many ways. Some examples are as follows:

- Friction between the soles of our shoes and the ground makes it possible for us to walk. Friction prevents us from slipping on the floor while walking.
- The friction between the tyres of vehicles and the road helps the tyres move forward as it helps to maintain a grip on the road.
- It helps us to stop moving vehicles, by applying brakes.
- It helps us to light a matchstick when we rub it on the rough surface of the matchbox.
- The friction between the lead of a pencil and the paper helps us to write.
- Friction between an eraser and paper helps us to erase pencil marks from the paper.

Friction between the soles of our shoes and the ground helps us to walk.

Friction between an eraser and paper

Negative Effects of Friction

Friction is not always useful for us. Some of the cases where friction can have negative effects are as follows:

- Friction causes wear and tear of the surfaces of objects that constantly rub against each other. For example, a pencil lead and paper, an eraser and paper, the soles of our shoes and the road, and the parts of machines.
- Friction gives rise to unwanted heat especially in vehicles. To reduce this heat in moving vehicles, coolants are used.

Fact File

When we rub our palms, they become warmer due to friction that generates the heat.

Streamlined shapes

- Some vehicles such as aeroplanes have a streamlined shape to reduce the effect of friction while moving at a high speed through air. Otherwise, they would burn a lot of fuel to move fast.
- Fish and birds have a streamlined body to reduce the effect of friction while moving in water and air, respectively. Otherwise, it would have been difficult for them to maintain balance while swimming or flying.

Case Study

Many Indian rural households still use grindstones to make coriander chutney and sharpen the blades of different tools. Frictional force comes into action when the two surfaces (the grindstone and coriander) are in contact. The force generates heat, which helps in the proper grinding and mixing of coriander leaves and the other added ingredients.

Based on the above information, answer the following questions:

a. Which force helps to make coriander chutney from its leaves?
b. Is the force discussed in the case above a contact force?
c. Give one more example of the useful effect of the force used above.

Wrap Up

- The force that acts between a moving object and the surface, which either slows down the object or stops it completely, is called the force of friction.
- The force of friction always opposes the force causing the movement of an object.

- Friction is useful in our day-to-day life. Without friction, no moving object will ever stop. In case there is too much friction, nothing will move.
- Friction is reduced by using substances that can smoothen out the two surfaces in contact with each other.

Exercises

SECTION I

REMEMBERING UNDERSTANDING

A Choose the correct option.

1. Which of the following can make it easier to slide a heavy box across the floor?
   a. Powder b. Oil c. Grease d. All of these
2. Why do we add powder on a carrom board?
   a. To increase friction b. To decrease friction
   c. To generate heat d. To generate light
3. Friction is more on a ..... surface.
   a. smooth b. rough c. plain d. none of these
4. ..... is applied on the door hinges to reduce friction.
   a. Oil b. Powder c. Wax d. Grease
5. You are able to walk due to ..... .
   a. powder b. oil c. grease d. friction
6. Aeroplanes have a ..... shape to reduce the effect of friction.
   a. hatchback b. broad
   c. streamlined d. non-streamlined

B Assertion and Reasoning questions.

1. **Assertion (A):** Oil is applied on the door hinges.
   Reason (R): Oil helps in reducing the creaking noise while opening or closing a door.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True

2. **Assertion (A):** Aeroplanes have streamlined body.

   Reason (R): It helps to reduce the effect of friction while moving at a high speed of air.

   a. Both A and R are True
   b. Both A and R are False
   c. A is True and R is False
   d. A is False and R is True

C **Choose the correct word to fill in the blank.**

1. Friction acts at the point of ..... (contact/rest).
2. Friction helps us to ..... (light/put off) a matchstick.
3. Frictional force is always in the direction ..... (similar/opposite) to that of a moving object.
4. The force between two surfaces in contact is called ..... (friction/lubricant).
5. ..... (Lubricant/Sand) is a substance that reduces friction by smoothening out the surfaces.
6. We can walk because there is ..... (friction/lubrication) between our feet and ground.

D **Write *T* for True and *F* for False. Correct the False statements.**

1. Friction makes it possible for a car to stop.
2. On a rough surface, a ball will roll to a greater distance than on a smooth surface.
3. Friction causes no damage to tyres, soles of shoes, or machine parts.
4. Friction is useful as well as harmful for us.
5. Friction should be reduced in engines.
6. Friction generates unwanted heat.

SECTION II

E **Short answer questions.**

1. What is friction?
2. Give two examples of friction related to our daily life.
3. Why is grease used in engines?

F **Long answer questions.**

1. What will happen if there is no friction? Explain giving suitable examples.
2. When trains run on tracks, it causes wear and tear on both surfaces. In worst cases, if either the tracks or wheels of the train are worn out, serious accidents may occur. Give reasons for this statement.
3. In amusement parks, it is easier to slide down on a polished slide than on a rough slide. Explain the possible reason for this.

Picture Study

Look at the given picture of four friends: Rakesh, Komal, Harshita, and Raghav. They are playing an indoor game. After observing the picture, answer the following questions.

1. Can you name the game that these four friends are playing?

   .....

2. What do the players need to apply in order to move the striker from its place?

   .....

3. If the carrom board is rough, will the striker move easily? Give a reason.

   .....

4. What can the players do if they want to move the striker easily?

   .....

My Learning Corner

SDG VALUES TECH INTEGRATION LIFE SKILLS CREATING

A Think about

Can you suggest why?

1. During winters, it is very difficult to walk on the snow-covered mountains.
2. Tyres of trucks and cars have grooved designs, and their surfaces are somewhat rough.

B Try out

TP

1. Suppose you visit a railway platform that is not clean. You see garbage and spilled water on the platform floor. Can you walk easily on such a floor? Create a poster using any software on your computer advising people on how to keep the platform clean.
2. Make a list of different situations from your day-to-day life where you experience friction. Also, suggest some measures to reduce them too.

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• define force due to friction.	
• explain how we can reduce friction.	
• explain the uses of friction.	
• explain the negative effects of friction.	

Worksheet

Complete the 'Friction' table given below.

1.	Define friction (in your own words).	....
2.	Give any two examples of friction from our daily life.	....
3.	Name any two ways in which we can reduce friction.	....
4.	Define a lubricant.	....
5.	Paste pictures of any two lubricants.	
6.	Write any two uses of friction.	....
7.	Write any two negative effects of friction.	....